Even and Odd Functions

Even

f is **even** when $f(-x) = f(x)$ for every x in its domain. Its graph is symmetric about the y-axis.

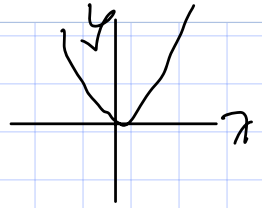
Odd

f is **odd** when $f(-x) = -f(x)$ for every x in its domain. Its graph is symmetric about the origin.

To classify a formula, compute $f(-x)$, simplify completely, and compare it with both $f(x)$ and $-f(x)$. A function may be neither even nor odd.

- if f is an even function, then when we plug in (input) $-x$, we get $f(x)$

ex: $f(x) = x^2$, $f(-x) = (-x)^2 = -x \cdot -x = x^2 = f(x)$



- if f is an odd function, input $-x$, output $-f(x)$,

ex. $f(x) = x^3$, plug in $-x$, $f(-x) = (-x)^3 = -x \cdot -x \cdot -x = x^2 \cdot -x = -x^3$

$= -f(x)$

Classify Each Function

Determine whether each function is **even**, **odd**, or **neither**. Show the algebraic test. plug in $-x$.

a. $f(x) = x^4 - 3x^2 + 1$

b. $g(x) = x^5 - 4x$

c. $h(x) = x^3 + x^2$

a. $f(x) = x^4 - 3x^2 + 1$, $f(-x) = (-x)^4 - 3(-x)^2 + 1 = x^4 - 3x^2 + 1 = f(x)$

even.

$(-x)^4 = -x \cdot -x \cdot -x \cdot -x = x \cdot x \cdot x \cdot x = x^4$

$(-x)^2 = -x \cdot -x = x^2$

$-4 \cdot -x = 4x$

b. $g(x) = x^5 - 4x$, plug in $-x$, $g(-x) = (-x)^5 - 4(-x)$

$= -x^5 + 4x = -g(x)$

$-g(x) = -x^5 + 4x$

(c) $h(x) = x^3 + x^2$

$-h(x) = -x^3 - x^2$

plug in $-x$, $h(-x) = (-x)^3 + (-x)^2$

$= -x^3 + x^2 \neq h(x)$

$\neq -h(x)$

neither.

Where Is the Function Increasing or Decreasing?

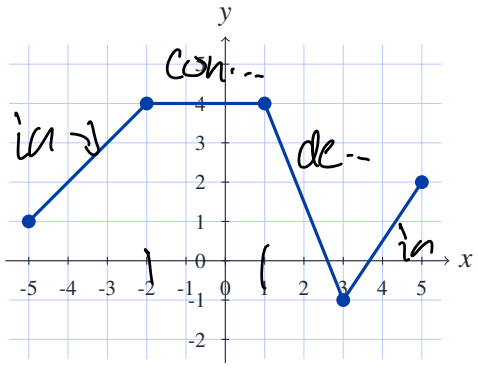
From the graph, state the intervals on which f is:

a. increasing, $(-5,-2) \cup (3,5)$

b. decreasing, $(1,3)$

c. constant. $(-2,2)$

Use interval notation and do not list isolated endpoints as intervals.

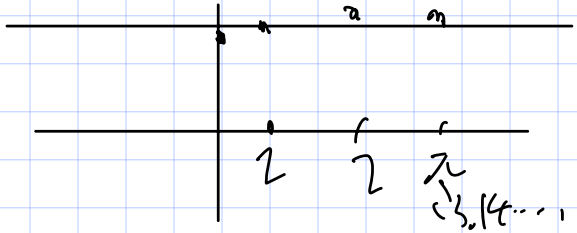


- constant. $f(x) = C$

ex: $f(x) = 6$

$f(-1) = 6$, $f(0) = 6$ .....

$f(x) = 6$

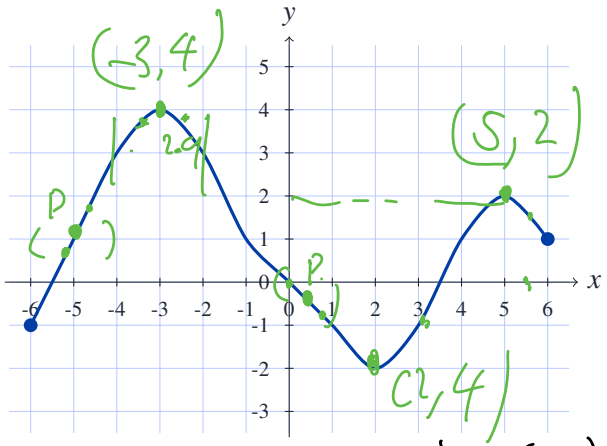


Identify Local and Absolute Extrema

Use the graph below. List every local maximum and minimum, then identify the absolute maximum and minimum on the shown domain. Report each as a point $(x, f(x))$.

plug in -3, $f(-3)=4$

$f(-2.9)$, $f(-3.2)$



$f(5.5)<2$

$f(3)<2$

- Local maximum:

A function has a local maximum at $x=a$, if $f(a)$ is greater than or equals to the functions at nearby x-values.

- absolute maximum.

A function has a absolute maximum at $x=a$, if $f(a)$ is greater than or equal to every function value

※ absolute is Largest local maximum.

Average Rate of Change

From $x = a$ to $x = b$, the average rate of change is $\dfrac{f(b) - f(a)}{b - a}$. It is also the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

For $f(x) = 2x^2 - 3x + 1$, find the average rate of change from $x = -1$ to $x = 2$. Interpret the sign of your answer.

- a, b are numbers, f(a), f(b) numbers, so this formula will give us a number.
- secant line is a line passes through 2 points on the graph of a function.
- slope is Average Rate of Change $= \dfrac{f(b)-f(a)}{b-a}$
- Give a function, $(-1, f(-1))$, $(2, f(2))$.

$f(-1) = 2\cdot(-1)^2 - 3(-2) + 1$

$= 2 + 3 + 1 = 6$

$f(2) = 2\cdot 2^2 - 3\cdot 2 + 1$

$= 8 - 6 + 1$

$= 3$

$\dfrac{f(b)-f(a)}{b-a}$

$= \dfrac{3-6}{2-(-1)} = \dfrac{-3}{3} = -1$

Find the Secant Line

Let $f(x) = x^2 - 4x + 5$. Find the equation of the secant line through the points on the graph with $x = 0$ and $x = 3$.

a. Find both points.

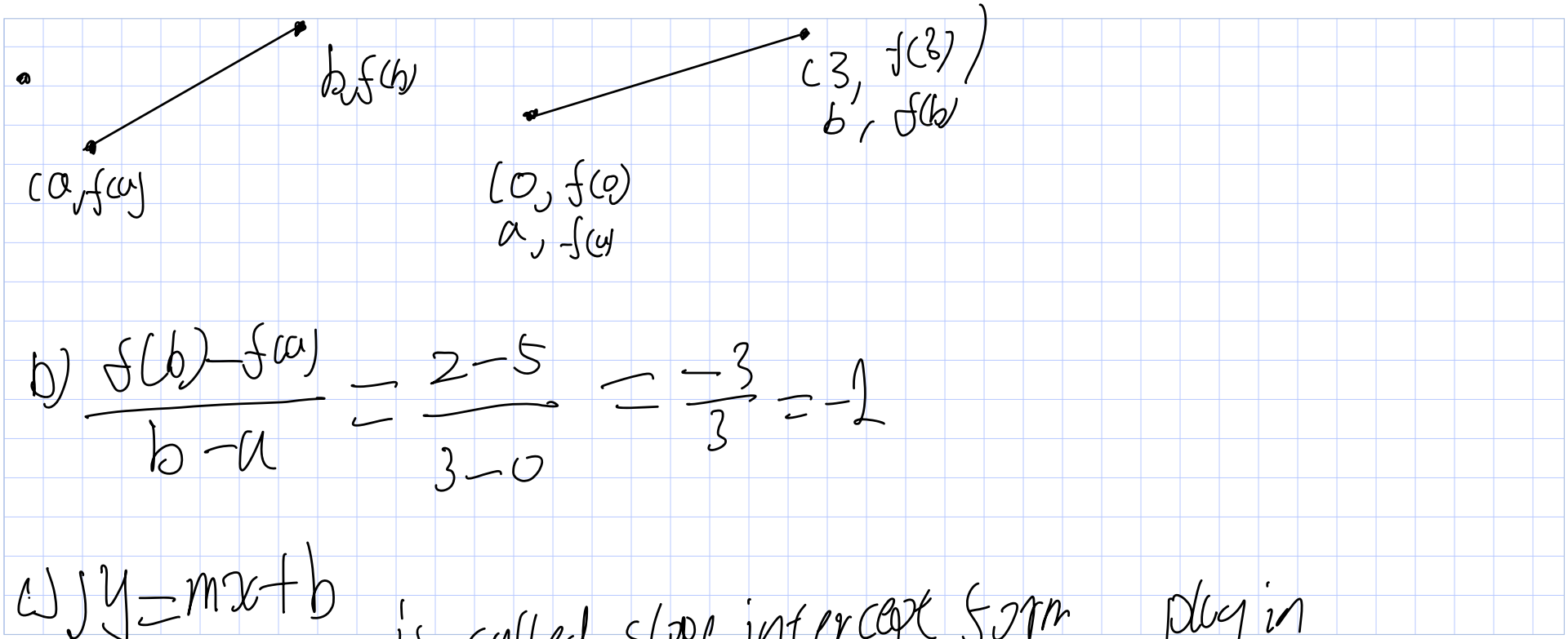
b. Compute the secant slope.

c. Write the line in slope-intercept form.

d. Check that both points satisfy your equation.

(a) find $f(0)$, $f(3)$

$= 5$ $\quad = 2$



b) $\dfrac{f(b)-f(a)}{b-a} = \dfrac{2-5}{3-0} = \dfrac{-3}{3} = -1$

c) $\begin{cases} y = mx+b \\ f(x) = mx+b \end{cases}$ is called slope intercept form

plug in $(0, 5)$

$y = -x+b$

$5 = -0+b$

m is slope or A.R.C.

$b=5$

$$\begin{cases} y=-x+5 \\ f(x)=-x+5 \end{cases}$$

$$\begin{cases} m>0, & \text{function is increasing,} \\ m=0, & \text{constant function,} \\ m<0, & \text{function is decreasing,} \end{cases}$$